

Extreme least singular values of Gaussian row submatrices and a phase retrieval stability problem

Yitzchak Shmalo^a

^a*Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Givat Ram,
Jerusalem, Israel*

Abstract

Let $\mathbb{F}$ be either $\mathbb{R}$ or $\mathbb{C}$, and put $d_{\mathbb{F}} = \dim_{\mathbb{R}} \mathbb{F}$. Let $A_m \in \mathbb{F}^{N_m \times m}$ have independent standard Gaussian entries and assume $N_m/m \rightarrow \gamma > 1$. We determine the exponential scale of the smallest least singular value among all square row submatrices:

$$M_m^{\mathbb{F}} = \min_{T \subset [N_m], |T|=m} \sigma_{\min}(A_{m,T}).$$

With $c_{\gamma} = \gamma^{\gamma}/(\gamma - 1)^{\gamma-1}$ we prove $\frac{1}{m} \log M_m^{\mathbb{F}} \xrightarrow{P} -d_{\mathbb{F}}^{-1} \log c_{\gamma}$, and when $N_m = \gamma m + O(1)$ the convergence holds at rate $O(m^{-1})$. As an application, for real phase retrieval at the critical number $N = 2m - 1$ of measurements the Balan–Wang stability parameter satisfies $\omega(A_m) = 4^{-m+o_P(m)}$, so the sharp exponential base in the Gaussian Balan–Wang problem is $1/4$. The lower estimate extends to all bounded–density ensembles, and continuity is necessary.

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1. Introduction

The least singular value of a random matrix is a basic hard-edge observable in random matrix theory. This paper studies a related extremal quantity: the minimum of the least singular value over all square row submatrices of

Email address: yitzchak.shmalo@gmail.com (Yitzchak Shmalo)

a tall Gaussian matrix. The number of such submatrices is exponential in the dimension, and the main point is to show that their strong dependence does not change the exponential scale predicted by the one-matrix hard-edge small-ball exponent.

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $d_{\mathbb{F}} = \dim_{\mathbb{R}} \mathbb{F}$, so that $d_{\mathbb{R}} = 1$ and $d_{\mathbb{C}} = 2$. A standard Gaussian variable over $\mathbb{R}$ is $N(0, 1)$; a standard Gaussian variable over $\mathbb{C}$ is $(\xi + i\eta)/\sqrt{2}$, where ξ, η are independent $N(0, 1)$ variables. For $A \in \mathbb{F}^{N \times m}$ and $T \subset [N]$, A_T denotes the submatrix formed from the rows indexed by T . Define

$$M_m^{\mathbb{F}}(A) = \min_{T \subset [N], |T|=m} \sigma_{\min}(A_T). \quad (1)$$

Our main result identifies the limiting exponential base for (1) for every fixed row aspect ratio $\gamma > 1$.

Theorem 1.1 (Extremal square submatrices). *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and $d_{\mathbb{F}} = \dim_{\mathbb{R}} \mathbb{F}$. Let $A_m \in \mathbb{F}^{N_m \times m}$ have independent standard Gaussian entries, and assume $N_m/m \rightarrow \gamma > 1$. Put*

$$h(\gamma) = \gamma \log \gamma - (\gamma - 1) \log(\gamma - 1), \quad c_{\gamma} = e^{h(\gamma)} = \frac{\gamma^{\gamma}}{(\gamma - 1)^{\gamma-1}}. \quad (2)$$

Then

$$\frac{1}{m} \log M_m^{\mathbb{F}}(A_m) \xrightarrow{P} -\frac{h(\gamma)}{d_{\mathbb{F}}}. \quad (3)$$

Equivalently,

$$M_m^{\mathbb{F}}(A_m) = c_{\gamma}^{-m/d_{\mathbb{F}} + o_P(m)}. \quad (4)$$

If in addition $N_m = \gamma m + O(1)$, for instance $N_m = \lfloor \gamma m \rfloor$, then for every fixed $\varepsilon > 0$,

$$\mathbb{P} \left\{ \left| \frac{1}{m} \log M_m^{\mathbb{F}}(A_m) + \frac{h(\gamma)}{d_{\mathbb{F}}} \right| > \varepsilon \right\} = O_{\gamma, \varepsilon, \mathbb{F}}(m^{-1}). \quad (5)$$

The constant c_{γ} is the exponential growth rate of the relevant binomial coefficient: $\binom{N_m}{m} = \exp\{h(\gamma)m + o(m)\}$. For one $m \times m$ Gaussian matrix over $\mathbb{F}$, the probability that $\sigma_{\min}$ is at most t is of order $t^{d_{\mathbb{F}}}$ at the origin, up to powers of m . If the square submatrices behaved independently, the minimum over $\exp\{h(\gamma)m + o(m)\}$ of them would therefore be expected at

scale $\exp\{-h(\gamma)m/d_{\mathbb{F}} + o(m)\}$. Theorem 1.1 proves that this heuristic gives the correct exponential scale.

A closely related question was studied by Rademacher and Shu [13] in the smoothed analysis of Frank–Wolfe methods. In our row notation, their results show that, when the number of Gaussian rows is a fixed factor larger than the dimension, some square submatrix has exponentially small least singular value, and all square submatrices have least singular value bounded below at a possibly different exponential scale. The present paper refines that picture by identifying the exact exponential rate, by treating both real and complex Gaussian matrices, and by extracting the sharp Gaussian base in the Balan–Wang phase retrieval stability question.

The motivating application is the stability of real finite-dimensional phase retrieval; this is the setting of Randomstrasse Open Problem 21. If $A \in \mathbb{R}^{N \times m}$ has rows $a_1, \dots, a_N$, the real phaseless measurement map is $x \bmod \{\pm 1\} \mapsto |Ax| = (|\langle a_i, x \rangle|)_{i=1}^N$. The finite-dimensional frame formulation goes back to Balan, Casazza and Edidin [2]; see also [7, 8, 12] for injectivity and stability background. In the real case, injectivity is characterized by the complement property: for every partition of the rows, one side must span $\mathbb{R}^m$. Thus the sharp generic threshold is $N = 2m - 1$.

Balan and Wang introduced quantitative stability parameters measuring the conditioning of spanning subfamilies and related these parameters to Lipschitz bounds for phaseless inversion [3]; see also [1, 4, 10]. At the real critical threshold, for a full-spark $A \in \mathbb{R}^{(2m-1) \times m}$, define

$$\omega(A) = \min_{J \subset [2m-1]: \text{rank}(A_J) < m} \sigma_m(A_{J^c}). \quad (6)$$

Balan and Wang conjectured an exponential upper bound

$$\omega(A) \leq C \max_i \|a_i\| b^m \quad (7)$$

with universal constants $C > 0$ and $0 < b < 1$. Randomstrasse Open Problem 21 asks for the value of $\omega(A)$ for iid Gaussian A and for the corresponding value of the exponential base in (7) [5, 6]. Thus the real critical case below is not just an example of Theorem 1.1; it is the stated Gaussian Balan–Wang open problem.

For full-spark $A \in \mathbb{R}^{(2m-1) \times m}$, (6) reduces to the minimum over square row submatrices:

$$\omega(A) = \min_{T \subset [2m-1], |T|=m} \sigma_{\min}(A_T). \quad (8)$$

Indeed, the rank-deficient sets J are exactly those with $|J| \leq m - 1$, and adding rows can only increase the Gram matrix. Applying Theorem 1.1 with $\mathbb{F} = \mathbb{R}$ and $N_m = 2m - 1$ gives the following answer.

Corollary 1.2 (Gaussian Balan–Wang base). *Let $A_m \in \mathbb{R}^{(2m-1) \times m}$ have independent standard Gaussian entries. Then*

$$\frac{1}{m} \log \omega(A_m) \xrightarrow{P} -\log 4, \quad (9)$$

with the rate

$$\mathbb{P} \left\{ \left| \frac{1}{m} \log \omega(A_m) + \log 4 \right| > \varepsilon \right\} = O_\varepsilon(m^{-1}) \quad (10)$$

for every fixed $\varepsilon > 0$. Equivalently, $\omega(A_m) = 4^{-m+o_P(m)}$.

Let $R_m = \max_{1 \leq i \leq 2m-1} \|a_i\|$. For every $b > 1/4$, $\mathbb{P}\{\omega(A_m) \leq R_m b^m\} \rightarrow 1$. For every $b < 1/4$ and every fixed $C < \infty$, $\mathbb{P}\{\omega(A_m) \leq C R_m b^m\} \rightarrow 0$. Thus the Gaussian critical base in the Balan–Wang question is $1/4$.

The complex part of Theorem 1.1 is not a statement about the complex phase retrieval threshold. It is the corresponding square-submatrix random matrix theorem over $\mathbb{C}$. For example, when $N_m/m \rightarrow 2$ one obtains $M_m^{\mathbb{C}}(A_m) = 2^{-m+o_P(m)}$, reflecting the two-real-dimensional small-ball exponent of a complex Gaussian scalar.

2. One-matrix lower tails and binomial growth

We first record the elementary one-matrix estimate needed for the lower bound. It is deliberately weaker than the sharp hard-edge asymptotics of Edelman [9] and subsequent work such as [14, 16]; the polynomial loss is harmless at the exponential scale considered here.

Lemma 2.1 (Polynomial least-singular-value tail). *Let $G \in \mathbb{F}^{m \times m}$ have independent standard Gaussian entries, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and $d_{\mathbb{F}} = \dim_{\mathbb{R}} \mathbb{F}$. There is a constant $C_{\mathbb{F}}$ such that, for all $m \geq 1$ and all $t > 0$,*

$$\mathbb{P}\{\sigma_{\min}(G) \leq t\} \leq C_{\mathbb{F}} m^{1+d_{\mathbb{F}}/2} t^{d_{\mathbb{F}}}. \quad (11)$$

Proof. Let $R_1, \dots, R_m$ be the rows of G , let $H_i = \text{span}_{\mathbb{F}}\{R_j : j \neq i\}$, and set $d_i = \text{dist}(R_i, H_i)$. Almost surely G is invertible. The negative second moment identity gives

$$\sum_{k=1}^m \sigma_k(G)^{-2} = \sum_{i=1}^m d_i^{-2}. \quad (12)$$

For completeness, if $y_i = G^{-1}e_i$, then $y_i \perp H_i$ and $\langle R_i, y_i \rangle = 1$, hence $\|y_i\| = d_i^{-1}$. Summing $\|y_i\|^2 = e_i^*(GG^*)^{-1}e_i$ over i proves (12).

If $\sigma_{\min}(G) \leq t$, then the left side of (12) is at least t^{-2} , so some i satisfies $d_i \leq \sqrt{m}t$. Conditional on the other rows, d_i has the same distribution as $|g_{\mathbb{F}}|$, where $g_{\mathbb{R}} \sim N(0, 1)$ and $g_{\mathbb{C}}$ is a standard complex Gaussian; hence $\mathbb{P}\{|g_{\mathbb{R}}| \leq u\} \leq \sqrt{2/\pi}u$ and $\mathbb{P}\{|g_{\mathbb{C}}| \leq u\} = 1 - e^{-u^2} \leq u^2$. A union bound over i gives (11). $\square$

We also use the following elementary asymptotic form of Stirling's formula.

Lemma 2.2 (Binomial scale). *Let $N_m/m \rightarrow \gamma > 1$. Then $\frac{1}{m} \log \binom{N_m}{m} \rightarrow h(\gamma)$, with h as in (2). If $N_m = \gamma m + O(1)$, then $\binom{N_m-1}{m-1} = \Theta_{\gamma}(m^{-1/2}e^{h(\gamma)m})$.*

3. Lower bound: no square submatrix is much smaller

Proposition 3.1 (Lower estimate). *Under the assumptions of Theorem 1.1, for every fixed $\varepsilon > 0$,*

$$\mathbb{P} \left\{ M_m^{\mathbb{F}}(A_m) \leq \exp \left[- \left(\frac{h(\gamma)}{d_{\mathbb{F}}} + \varepsilon \right) m \right] \right\} \rightarrow 0. \quad (13)$$

If $N_m = \gamma m + O(1)$, the probability in (13) is bounded by $C_{\gamma, \varepsilon, \mathbb{F}} m^{1+d_{\mathbb{F}}/2} e^{-d_{\mathbb{F}}\varepsilon m}$ for all large m .

Proof. Let $t > 0$. By a union bound and Lemma 2.1, $\mathbb{P}\{M_m^{\mathbb{F}}(A_m) \leq t\} \leq \binom{N_m}{m} C_{\mathbb{F}} m^{1+d_{\mathbb{F}}/2} t^{d_{\mathbb{F}}}$. Take $t = \exp[-(h(\gamma)/d_{\mathbb{F}} + \varepsilon)m]$. Since $m^{-1} \log \binom{N_m}{m} \rightarrow h(\gamma)$, the bound tends to zero. The stated quantitative bound follows from Lemma 2.2 when $N_m = \gamma m + O(1)$. $\square$

4. Upper bound: one row and many random hyperplanes

The matching upper estimate is the main point. We fix one row and compare it to all hyperplanes generated by $m-1$ of the remaining rows. Let

A_m have rows $a_1, \dots, a_N$, $N = N_m$. Put $x = a_N$ and $b_i = a_i$ for $1 \leq i \leq N-1$, and define

$$\mathcal{S} = \{S \subset [N-1] : |S| = m-1\}, \quad L = |\mathcal{S}| = \binom{N-1}{m-1}.$$

For $S \in \mathcal{S}$, let $H_S = \text{span}_{\mathbb{F}}\{b_i : i \in S\}$, a.s. a hyperplane in $\mathbb{F}^m$. Choose a unit normal u_S to H_S (over $\mathbb{C}$ determined up to a phase; all events below are phase-invariant). For $t > 0$ set

$$X_S = \mathbf{1}_{\{|\langle x, u_S \rangle| \leq t\}}, \quad Z_t = \sum_{S \in \mathcal{S}} X_S. \quad (14)$$

If $Z_t > 0$, then for some S the row x lies within distance t of H_S ; for $T = S \cup \{N\}$ and a unit vector $v \perp H_S$, $\|A_{m,T}v\| = |\langle x, v \rangle| = \text{dist}(x, H_S) \leq t$. Thus

$$Z_t > 0 \implies M_m^{\mathbb{F}}(A_m) \leq t. \quad (15)$$

4.1. Pair geometry

Lemma 4.1 (Normals for two overlapping row sets). *Fix distinct $S, T \in \mathcal{S}$. Let $r = |S \cap T|$ and $D = m - r$. Then $D \geq 2$. Conditional on the rows indexed by $S \cap T$, the two normal lines $\mathbb{F}u_S$ and $\mathbb{F}u_T$ are independent Haar-distributed lines in the D -dimensional space $E^\perp$, $E = \text{span}_{\mathbb{F}}\{b_i : i \in S \cap T\}$. Equivalently, for all phase-invariant events one may take u_S, u_T to be independent Haar unit vectors in $E^\perp$.*

Proof. Almost surely $\dim_{\mathbb{F}} E = r$, so $\dim_{\mathbb{F}} E^\perp = D$. The sets $S \setminus T$ and $T \setminus S$ each contain $D-1$ rows, and the corresponding row families are independent conditional on the rows in $S \cap T$. Projecting the rows in $S \setminus T$ to $E^\perp$ gives $D-1$ independent standard Gaussian vectors in $E^\perp$; their span is a hyperplane, and by rotational invariance its normal line is Haar-distributed. The same argument applies to $T \setminus S$, independently. $\square$

For $D \geq 2$ define, with u, v independent Haar unit vectors in $\mathbb{F}^D$ and g an independent standard Gaussian vector in $\mathbb{F}^D$,

$$q_D^{\mathbb{F}}(t) = \mathbb{P}\{|\langle g, u \rangle| \leq t, |\langle g, v \rangle| \leq t\}, \quad p_{\mathbb{F}}(t) = \mathbb{P}\{|g_{\mathbb{F}}| \leq t\}, \quad (16)$$

where $g_{\mathbb{F}}$ is a standard scalar Gaussian over $\mathbb{F}$.

Lemma 4.2 (Two-slab and two-disk estimate). *For each $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ there is a constant $C_{\mathbb{F}}$ such that, for $0 < t \leq 1$,*

$$q_D^{\mathbb{F}}(t) \leq (1 + C_{\mathbb{F}} t^2) p_{\mathbb{F}}(t)^2 \left(1 + \frac{C_{\mathbb{F}}}{D}\right), \quad D \geq 3, \quad (17)$$

and $q_2^{\mathbb{F}}(t) \leq p_{\mathbb{F}}(t)$.

Proof. The case $D = 2$ is immediate, since the joint event is contained in $\{|\langle g, u \rangle| \leq t\}$. Assume $D \geq 3$. For $\mathbb{F} = \mathbb{R}$, conditional on u, v write $\rho = \langle u, v \rangle$ and $s = (1 - \rho^2)^{1/2}$; the pair $(\langle g, u \rangle, \langle g, v \rangle)$ is a centered real Gaussian vector with covariance determinant s^2 , density at the origin $(2\pi s)^{-1}$, so

$$\mathbb{P}\{|\langle g, u \rangle| \leq t, |\langle g, v \rangle| \leq t \mid u, v\} \leq \frac{2t^2}{\pi s}. \quad (18)$$

For independent Haar unit vectors in $\mathbb{R}^D$, $\mathbb{E}(1 - \rho^2)^{-1/2} = \Gamma(D/2)\Gamma((D - 2)/2)/\Gamma((D - 1)/2)^2 \leq 1 + C/D$ by the gamma-ratio asymptotics. Averaging (18) and using $p_{\mathbb{R}}(t) = \sqrt{2/\pi} t + O(t^3)$ gives (17) over $\mathbb{R}$. For $\mathbb{F} = \mathbb{C}$, conditional on u, v write $s^2 = 1 - |\rho|^2$ with $\rho = \langle u, v \rangle$; the pair is a centered complex Gaussian vector in $\mathbb{C}^2$ with covariance determinant s^2 and density at the origin $(\pi^2 s^2)^{-1}$, so the probability of the product of the two disks of radius t is at most $t^4/(1 - |\rho|^2)$. For independent Haar unit vectors in $\mathbb{C}^D$, $|\rho|^2$ has the Beta(1, $D - 1$) distribution, hence $\mathbb{E}(1 - |\rho|^2)^{-1} = (D - 1)/(D - 2) \leq 1 + C/D$ for $D \geq 3$. Since $p_{\mathbb{C}}(t) = 1 - e^{-t^2} = t^2 + O(t^4)$, averaging gives (17) over $\mathbb{C}$. $\square$

Lemma 4.3 (Overlap combinatorics). *Let S, T be independent uniform elements of $\mathcal{S} = \{S \subset [N - 1] : |S| = m - 1\}$, and set $D = m - |S \cap T|$. Then for $d \geq 1$,*

$$\mathbb{P}\{D = d\} = \frac{\binom{m-1}{d-1} \binom{N-m}{d-1}}{\binom{N-1}{m-1}}, \quad (19)$$

with the convention that binomial coefficients outside their natural range vanish. Moreover

$$\mathbb{E} \frac{1}{D} = \frac{N}{m(N - m + 1)}, \quad \mathbb{P}\{D = 2\} = \frac{(m - 1)(N - m)}{\binom{N-1}{m-1}}. \quad (20)$$

Proof. Fix S . To have $D = d$, the set T omits $d - 1$ elements of S and includes $d - 1$ elements from $[N - 1] \setminus S$ (of size $N - m$); this gives (19). With $n = N - 1$ and $k = m - 1$,

$$\mathbb{E} \frac{1}{D} = \frac{1}{\binom{n}{k}} \sum_{j=0}^k \frac{1}{j+1} \binom{k}{j} \binom{n-k}{j} = \frac{1}{\binom{n}{k}} \cdot \frac{1}{k+1} \sum_{j=0}^k \binom{k+1}{j+1} \binom{n-k}{j} = \frac{1}{\binom{n}{k}} \cdot \frac{1}{k+1} \binom{n+1}{k},$$

using $(j+1)^{-1} \binom{k}{j} = (k+1)^{-1} \binom{k+1}{j+1}$ and Vandermonde's identity. Hence $\mathbb{E}[1/D] = \frac{n+1}{(k+1)(n+1-k)} = \frac{N}{m(N-m+1)}$. The value of $\mathbb{P}\{D = 2\}$ is (19) at $d = 2$. $\square$

4.2. The second moment

Proposition 4.4 (Upper estimate). *Under the assumptions of Theorem 1.1, for every fixed $\varepsilon > 0$,*

$$\mathbb{P} \left\{ M_m^{\mathbb{F}}(A_m) > \exp \left[- \left(\frac{h(\gamma)}{d_{\mathbb{F}}} - \varepsilon \right) m \right] \right\} \rightarrow 0. \quad (21)$$

If $N_m = \gamma m + O(1)$, the probability in (21) is $O_{\gamma, \varepsilon, \mathbb{F}}(m^{-1})$.

Proof. Take $0 < \varepsilon < h(\gamma)/d_{\mathbb{F}}$. First assume $N_m = \gamma m + O(1)$, and write $h = h(\gamma)$, $d = d_{\mathbb{F}}$, $t = \exp[-(h/d - \varepsilon)m] \in (0, 1]$ for large m . With Z_t as in (14) and $L = \binom{N-1}{m-1}$, $\mu := \mathbb{E}Z_t = L p_{\mathbb{F}}(t)$. Since $p_{\mathbb{F}}(t) \geq c_{\mathbb{F}} t^d$ for $0 < t \leq 1$ and $L \geq c_{\gamma} m^{-1/2} e^{hm}$ (Lemma 2.2),

$$\mu \geq c_{\gamma, \mathbb{F}} m^{-1/2} e^{d\varepsilon m} \rightarrow \infty. \quad (22)$$

For $S \neq T$, Lemma 4.1 and the independence of x from all b_i give $\mathbb{E}[X_S X_T] = q_{D(S,T)}^{\mathbb{F}}(t)$ with $D(S, T) = m - |S \cap T|$ (projecting x to $E^{\perp}$ makes it a standard Gaussian vector there, independent of u_S, u_T). Hence, for S, T independent uniform in $\mathcal{S}$ and $D = m - |S \cap T|$,

$$\frac{\mathbb{E}Z_t^2}{\mu^2} \leq \frac{1}{L p_{\mathbb{F}}(t)} + \mathbb{E} \left[\frac{q_D^{\mathbb{F}}(t)}{p_{\mathbb{F}}(t)^2} \mathbf{1}_{\{D \geq 2\}} \right]. \quad (23)$$

By Lemmas 4.2 and 4.3, the $D \geq 3$ part is at most $(1 + C_{\mathbb{F}} t^2)(1 + C_{\mathbb{F}} \mathbb{E}[1/D]) \leq 1 + C_{\mathbb{F}} t^2 + C_{\gamma, \mathbb{F}}/m$ (using $\mathbb{E}[1/D] = O(1/m)$), and the $D = 2$ part is at most $\mathbb{P}\{D = 2\}/p_{\mathbb{F}}(t) \leq C_{\gamma, \mathbb{F}} m^{5/2} e^{-d\varepsilon m}$. With (22),

$$\frac{\text{Var } Z_t}{\mu^2} = \frac{\mathbb{E}Z_t^2}{\mu^2} - 1 \leq \frac{C_{\gamma, \mathbb{F}}}{m} + C_{\gamma, \varepsilon, \mathbb{F}} m^{5/2} e^{-d\varepsilon m} = O_{\gamma, \varepsilon, \mathbb{F}}(m^{-1}),$$

so by Chebyshev $\mathbb{P}\{Z_t = 0\} \leq \text{Var}(Z_t)/\mu^2 = O(m^{-1})$. With (15) this proves (21) and its rate when $N_m = \gamma m + O(1)$. For a general sequence $N_m/m \rightarrow \gamma$, repeat the proof with $h(N_m/m)$ in place of $h(\gamma)$ and use $h(N_m/m) \rightarrow h(\gamma)$. $\square$

5. Proofs of the main theorem and the phase retrieval corollary

Proof of Theorem 1.1. Propositions 3.1 and 4.4 give, for every $\varepsilon > 0$, that $\frac{1}{m} \log M_m^{\mathbb{F}}(A_m)$ is below $-h(\gamma)/d_{\mathbb{F}} - \varepsilon$ and above $-h(\gamma)/d_{\mathbb{F}} + \varepsilon$ each with probability $\rightarrow 0$; this is convergence in probability. When $N_m = \gamma m + O(1)$ the lower probability is exponentially small and the upper probability is $O(m^{-1})$, giving (5). $\square$

Proof of Corollary 1.2. For Gaussian $A_m \in \mathbb{R}^{(2m-1) \times m}$, every m rows are independent almost surely, so the reduction (8) holds a.s. Theorem 1.1 with $\mathbb{F} = \mathbb{R}$, $N_m = 2m - 1$ and $h(2) = \log 4$ gives (9)–(10). For the base in (7), note $\frac{1}{m} \log R_m \xrightarrow{P} 0$: the upper tail follows from a union bound and $\mathbb{P}\{\|g\| \geq e^{\eta m}\} \leq \exp(-ce^{2\eta m})$ for fixed $\eta > 0$, and $R_m \geq 1$ with high probability. If $b > 1/4$, pick ε with $e^{-(\log 4 - \varepsilon)} < b$; then w.h.p. $\omega(A_m) \leq e^{-(\log 4 - \varepsilon)m} \leq b^m \leq R_m b^m$. If $b < 1/4$ and $C < \infty$, pick ε, η with $e^{\eta b} < e^{-(\log 4 + \varepsilon)}$; then w.h.p. $\omega(A_m) \geq e^{-(\log 4 + \varepsilon)m}$ and $R_m \leq e^{\eta m}$, so $CR_m b^m \leq C(e^{\eta b})^m < e^{-(\log 4 + \varepsilon)m}$ for large m , whence $\mathbb{P}\{\omega(A_m) \leq CR_m b^m\} \rightarrow 0$. $\square$

Remark 5.1 (What drives the constants). *The theorem separates the two inputs that determine the exponential base. The combinatorial input is the entropy $\binom{N_m}{m} = \exp\{h(\gamma)m + o(m)\}$. The small-ball input is the real dimension of the scalar field: $\mathbb{P}\{|g_{\mathbb{R}}| \leq t\} \asymp t$ while $\mathbb{P}\{|g_{\mathbb{C}}| \leq t\} \asymp t^2$. Thus the same binomial entropy is divided by $d_{\mathbb{F}} = 1$ over $\mathbb{R}$ and by $d_{\mathbb{F}} = 2$ over $\mathbb{C}$.*

6. Universality and the necessity of continuity

The lower estimate is insensitive to the entry law, provided it has a bounded density.

Proposition 6.1 (Universal lower estimate). *Let $A_m \in \mathbb{R}^{N_m \times m}$ have i.i.d. entries with density bounded by K , and $N_m/m \rightarrow \gamma > 1$. Then for every fixed $\varepsilon > 0$, $\mathbb{P}\{M_m^{\mathbb{R}}(A_m) \leq e^{-(h(\gamma) + \varepsilon)m}\} \rightarrow 0$.*

Proof. The identity (12) is purely algebraic. For $G \in \mathbb{R}^{m \times m}$ with rows R_i and n_i the unit normal to $\text{span}\{R_j : j \neq i\}$, the variable $\langle R_i, n_i \rangle = \sum_j (R_i)_j (n_i)_j$ is, conditionally on the other rows, a sum of independent terms; the term with $|(n_i)_{j^*}| = \max_j |(n_i)_j| \geq 1/\sqrt{m}$ has density at most $K/|(n_i)_{j^*}| \leq K\sqrt{m}$, so the convolution has density at most $K\sqrt{m}$ and $\mathbb{P}\{d_i \leq s\} \leq 2K\sqrt{m}s$. Hence $\mathbb{P}\{\sigma_{\min}(G) \leq t\} \leq 2Km^2t$, and a union bound over $\binom{N_m}{m} = e^{h(\gamma)m+o(m)}$ blocks tends to 0 at $t = e^{-(h(\gamma)+\epsilon)m}$. $\square$

Remark 6.2 (Continuity is necessary). *For atomic entries the lower estimate fails. If the entries are ± 1 Bernoulli and $N = 2m - 1$, every $m \times m$ submatrix is singular with probability at least $\mathbb{P}\{R_1 = R_2\} = 2^{-m}$, so the expected number of singular $m \times m$ submatrices is at least $\binom{2m-1}{m} 2^{-m} \geq c 2^m / \sqrt{m} \rightarrow \infty$. Thus the first-moment heuristic behind Proposition 6.1 breaks down, and a bounded-density assumption (in particular Gaussianity) cannot be removed.*

Remark 6.3 (Towards a universal upper estimate). *The upper estimate is expected to persist for i.i.d. subgaussian entries with bounded density. The only use of Gaussianity is Lemma 4.1, where rotational invariance makes the hyperplane normals Haar-distributed. For bounded-density entries the normals are delocalized with high probability, so a density-level local central limit theorem yields $\mathbb{P}\{|\langle x, u_S \rangle| \leq t\} = \Theta(t)$ together with the matching pair estimate; making this uniform over the $\binom{N-1}{m-1}$ hyperplanes requires a delocalization bound we do not pursue here. A Berry–Esseen estimate does not suffice at the scale $t = e^{-\Theta(m)}$, since its $O(m^{-1/2})$ error dwarfs the signal; a density-level statement is needed.*

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools (large language model assistants) to assist with drafting, revision and bibliographic organization. After using these tools, the author reviewed and

edited the content as needed and takes full responsibility for the content of the manuscript.

Declaration of competing interest

The author declares no competing interests.

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